Hugo Marquerie 20/03/2026

Ejer desigualdad triangular producto hiperbolico

Ejercicio 1. Sean $x, y, z \in [0, 1)$ tales que $x \leqslant \dfrac{y+z}{1+yz}$. Demuestra que

$$\frac{1+x}{1-x} \leqslant \frac{1+y}{1-y} \cdot \frac{1+z}{1-z}.$$

***Solución*:** Consideramos la función $\varphi\colon \mathbb{R} \setminus \{1\} \to \mathbb{R}$ dada por $\varphi(t) = \frac{1+t}{1-t}$. Su derivada es $\varphi'(t) = \frac{2}{(1-t)^2} > 0$ para todo $t < 1$, por lo que φ es estrictamente creciente en $(-\infty, 1)$.

Por tanto, como $x, y, z \in [0, 1)$ y $x \leqslant \frac{y+z}{1+yz}$, tenemos que

$$\varphi(x) \leqslant \varphi\left(\frac{y+z}{1+yz}\right) \implies \frac{1+x}{1-x} \leqslant \frac{1 + \frac{y+z}{1+yz}}{1 - \frac{y+z}{1+yz}}.$$

Operando en el lado derecho, llegamos a que

$$\frac{1 + \frac{y+z}{1+yz}}{1 - \frac{y+z}{1+yz}} = \frac{(1+yz) + (y+z)}{(1+yz) - (y+z)} = \frac{(1+y)(1+z)}{(1-y)(1-z)} = \frac{1+y}{1-y} \cdot \frac{1+z}{1-z} = \varphi(y) \cdot \varphi(z).$$

Esto implica la desigualdad que queríamos demostrar. ■

Referenciado en

- `prop-metrica-poincare`

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.